Vipul Sapra

Kharghar, Navi Mumbai | 8888-033-168 | vpsapra5@gmail.com | linkedin.com/in/vipul-sapra-ab30a09a

EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Distributed Systems, Large-Scale Data Platforms, Predictive Modeling, Anomaly Detection, Dimensional Data Modeling, Star/Snowflake Schemas, Slowly Changing Dimensions (SCD Type 1/2), Automated Data Quality Governance, CI/CD, Operational Health & Monitoring, Unit Economics Simulation.

Programming Languages & Tools: AWS (S3, Redshift), Azure (Data Factory, Synapse, SQL, Databricks), Snowflake, Apache Spark, PySpark, XGBoost, Random Forests, Scikit-Learn, SARIMA, Power BI, Tableau, Airflow, One Automation, Python, SQL.

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Software Development Engineer II - Profit Intelligence & Large-Scale Data Systems — **May 2022 – Present**

- **Architected and engineered scalable petabyte-scale data platforms** utilizing high-performance SQL stored procedures, dimensional data modeling (Star/Snowflake schemas), and Slowly Changing Dimensions (SCD Type 1/2) to ingest and process massive financial and operational event streams with 99.9% data availability.
- **Built and operated foundational distributed data infrastructure** leveraging AWS (S3, Redshift), Azure Cloud (ADF, Synapse, Databricks), and Apache Spark / PySpark to crunch hundreds of terabytes of incoming multi-source data, ensuring fully configurable and self-service reliability for cross-functional business units.
- **Designed and deployed machine learning prediction engines** using XGBoost, Random Forests, and Scikit-Learn for customer behavior modeling, alongside advanced SARIMA forecasting models for capacity planning, driving a 20% improvement in system reliability and response times.
- **Integrated advanced AI and automated data quality firewalls** incorporating agentic AI, prompt engineering, and automated validation frameworks to perform real-time anomaly detection with 95% accuracy, reducing operational turnaround and data discrepancy remediation by over 80%.
- **Owned end-to-end production reliability and operational health** establishing Airflow and One Automation batch monitoring, rigorous Root Cause Analysis (RCA) postmortems, comprehensive runbooks, and automated QA testing frameworks that cut manual turnaround times by 50%.

Master of Science in Information Systems & Data Science — **May 2022 – Present**

- **Mastered advanced data structures and algorithms** through rigorous coursework at the University of Cincinnati, maintaining a 3.95/4.0 GPA while focusing on large-scale systems design.
- **Conducted applied machine learning and statistical modeling research** focusing on predictive analytics, distributed data processing, and enterprise database architectures.
- **Engineered end-to-end data pipelines and analytics solutions** bridging theoretical computer science concepts with practical software engineering applications.

Bachelor of Engineering in Petrochemical — **May 2022 – Present**

- **Built strong foundational analytical and mathematical problem-solving skills** at Pune University MIT, applying rigorous quantitative methods to complex system optimization.
- **Modeled complex physical and chemical process dynamics** utilizing simulation software and statistical tools to drive efficiency and operational improvements.
- **Collaborated on multidisciplinary engineering projects** emphasizing rigorous process control, safety standards, and structured technical documentation.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.

- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture **Pune, India**

Test Engineering Analyst **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.